

Research

Ancient dog introgression into the Iberian wolf genome may have facilitated adaptation to human-dominated landscapes

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Understanding how large carnivores respond to increasingly human-dominated landscapes will determine their future adaptive potential. The Iberian wolf (*Canis lupus signatus*), a gray wolf subspecies endemic to the Iberian Peninsula (Portugal and Spain), has uniquely persisted in human-dominated landscapes, unlike many other wolf populations that faced widespread extinction across Europe during the twentieth century. In this study, we conducted a comprehensive genome-wide analysis of 145 historical and contemporary Iberian wolf samples to investigate whether hybridization with domestic dogs resulted in genetic introgression. We identified a dog-derived block on Chromosome 2 in Iberian wolves, displaying signatures consistent with introgression and high nucleotide similarity among introgressed individuals. Additionally, our estimates place the average timing of introgression between 6100 and 3000 years ago, with low sequence divergence to dogs from the Iberian Peninsula suggesting a single local origin for the hybridization event. Using forward genetic simulations, we show that the introgressed haplotype is most likely being maintained in Iberian wolves by selection. The introgressed dog variants are located within the *MAST4* gene, which has been linked to neurological disorders, including cognitive and motor developmental delays, hinting at a potential role in cognitive behavior in Iberian wolves. This study uncovers a case of putative adaptive introgression from domestic dogs into wolves, offering new insights into wild canids' adaptation to human-dominated landscapes.

[Supplemental material is available for this article.]

Circa 10,000 years ago, humans started to change the planet, prompted by the advent of farming following the domestication of plants and animals (Frantz et al. 2020). Over time, large carnivores have been severely affected by these rapid environmental changes, mostly because of the increasing competition with humans (e.g., persecution mediated by livestock depredation), and many have gone (locally) extinct (Chapron et al. 2014). Notably, amid these challenges, some species managed to change their behavior and ecology to adapt to human activities in human-dominated landscapes (Benazzo et al. 2017). A remarkable example of such resilience is illustrated by some populations of the Eurasian gray wolf (*Canis lupus*) living in highly human-dominated land-

scapes (Sazatornil et al. 2016; Rio-Maior et al. 2019). Such landscapes are found in the Iberian Peninsula (i.e., Portugal and Spain), where the human population density ranges from 20 to 400 inhabitants per square kilometer in the wolf's distribution (Instituto Nacional de Estatística Portugal 2017) (vs. European average: 97 inhabitants per square kilometer) (Chapron et al. 2014).

The Iberian wolf subspecies (*Canis lupus signatus*) (Fig. 1C) diverged from other Eurasian wolves about 10,000 years ago and remains isolated (Silva et al. 2020). Despite the widespread extinction of wolf populations in Europe during the twentieth century (Mech and Boitani 2003), the Iberian wolf exhibits remarkable resilience regardless of intense human persecution (Sastre et al. 2011; Nores and López-Bao 2022). Notably, this subspecies not only persists in areas characterized by high human density but

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